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[20EC723]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER - 2023

OPEN SOURCE SYSTEMS

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	What are open source systems? Mention the need and advantages of such systems.	**06**
	b.	Explain the concepts of Pipes and Redirection with respect to Linux.	**08**
2.	**a.**	Write a shell script to display the use of while command.	**06**
	b.	Write a shell script to show working with all types of if-then statement.	**08**
3.	**a.**	What is meant by path and pathname in Linux? Explain them in detail.	**07**
	b.	Write about the operations unique to Linux directories alone.	**07**
		OR	
4.	**a.**	Discuss different types of files in Linux.	**07**
	b.	Distinguish between Home and Root Directory.	**07**
5.	**a.**	How users are managed in Linux? Explain all the available commands for the purpose.	**07**
	b.	What are the various file permissions which could be given? How rights could be given or revoked?	**07**
		OR	
6.	**a.**	Explain the concept of file compression in Linux.	**07**
	b.	Explain how Linux config files can be edited.	**07**
7.	**a.**	Enumerate the features and limitations of MySQL.	**06**
	b.	Consider the following tables DRESS and MATERIAL. Write SQL commands for the statements (i) to (iv) and give outputs for SQL queries (v) to (viii).	**08**

Table : DRESS

DCODE	DESCRIPTION	PRICE	MCODE	LAUNCHDATE
10001	FORMAL SHIRT	1250	M001	12–JAN–08
10020	FROCK	750	M004	09–SEP–07
10012	INFORMAL SHIRT	1450	M002	06–JUN–08
10019	EVENING GOWN	850	M003	06–JUN–08
10090	TULIP SKIRT	850	M002	31–MAR–07
10023	PENCIL SKIRT	1250	M003	19–DEC–08
10089	SLACKS	850	M003	20–OCT–08
10007	FORMAL PANT	1450	M001	09–MAR–08
10009	INFORMAL PANT	1400	M002	20–OCT–08
10024	BABY TOP	650	M003	07–APR–08

Table : MATERIAL

MCODE	TYPE
M001	TERELENE
M002	COTTON
M004	POLYESTER
M003	SILK

i. To display DCODE and DESCRIPTION of an each dress in ascending order of DCODE.

ii. To display the details of all the dresses which have LAUNCHDATE in between 05–DEC–07 and 20–JUN–08 (inclusive of both the dates).

iii. To display the average PRICE of all the dresses which are made up of material with MCODE as M003.

iv. To display material wise highest and lowest price of dresses from DRESS table. (Display MCODE of each dress along with highest and lowest price)

v. SELECT SUM (PRICE) FROM DRESS WHERE MCODE='M001';

vi. SELECT DESCRIPTION, TYPE FROM DRESS, MATERIAL WHERE DRESS. DCODE=MATERIAL. MCODE AND DRESS. PRICE>=1250;

vii. SELECT MAX(MCODE) FROM MATERIAL;

viii. SELECT COUNT(DISTINCT PRICE) FROM DRESS

OR

8. a. What is DBA? Explain roles and responsibilities of DBA. **06**

b. Write code for database connectivity. **08**